

Yusuf Afifi

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EDUCATION

UC Berkeley

Berkeley, CA

Masters of Engineering in Electrical Engineering and Computer Science

2026 – 2027

- Concentration in Visual Computing

University College London

London, UK

Bachelor of Science (Honours) in Computer Science, Minor in Applied Mathematics

2023 – 2026

- First Class Honours
- Teaching Assistant for COMP0002 in C, COMP0004 in Java, ENGF0034 in Python, and COMP0016 Systems
- Executive Quant Ventures at UCL Fintech Society

EXPERIENCE

AI Research Fellow | *RL, Post-training*

San Francisco, US

Future Principle

June 2026 – September 2026

- Post-trained open-source models in prediction market environments to enhance their predictive intelligence and reasoning capabilities.
- Curated a high-quality dataset of **8000** resolved binary and categorical events from Polymarket and Kalshi, building a verifiable evaluation environment via Prime Intellect.
- Generated training rollouts using Qwen **3.5 27B** within an agentic harness, integrating the Polymarket API and web search tools for model post-training.
- Benchmarked the post-trained model on targeted prediction benchmarks, beating the win rate of frontier models (Opus **4.8** and GPT-**5.6**) by at least **2%**.

Software Development Engineer Intern | *Gen AI, AWS, Cloud, AI Agent*

London, UK

Amazon Prime Video

June 2025 – September 2025

- Reduced ad-hoc query and report generation time by **75%** by deploying a text-to-SQL agentic AI solution for Prime Video SVOD accounting team. Engineered a **RAG** workflow using **AWS Bedrock** hosting **Claude** and a **Kendra** index (containing DB schemas and business logic) to ground the agent before executing queries on **Redshift**.
- Launched a production-grade internal web application, available Amazon-wide, featuring a **React** frontend and a **Lambda** proxy handling a **WebSocket API GW** for a chatbot interface. Automated infrastructure provisioning using CI/CD and **AWS CDK**, while securing all contract data with least-privilege **IAM** policies.
- Curated a complete **CloudWatch** observability stack with dashboards and automated alarms on performance and cost metrics (e.g., Bedrock token usage, Lambda latency), ensuring high availability for an application supporting **200+ weekly active users** across global accounting teams.

PROJECTS AND RESEARCH

🌸 **Petals: AI-powered Health & Wellness app** | *iOS Development, On-Device AI*

July 2025

- Published the **first** iOS app that leverages **Apple Foundation Models** for private, on-device inference on the Neural Engine, featuring a **RAG** framework that securely accesses **Apple HealthKit** data for context.
- Engineered an instruction-tuned **LLM** chatbot (“Petal”) that dynamically adapts its responses to user health and activity data, ensuring complete privacy through exclusive **on-device AI** processing.
- Powering a suite of AI-driven features—including personalized meditations, AI journaling, and curated wellness plans through advanced **context engineering** and feature integration with the foundation model.

🌸 **ReadingStar: Accessibility AI Karaoke Game** | *On-Device Voice AI, Intel, Windows App*

April 2025

- Led the development of a gamified Windows karaoke app with **Intel** & the **National Autistic Society** to aid speech development in neurodiverse students, achieving national school distribution (e.g. Helen Allison) and a feature by Intel as the first education app to run on an Intel NPU.
- Engineered a real-time, fully offline AI pipeline for speech-to-text, audio-to-lyric alignment, and accuracy scoring, optimized with lightweight models for low-latency inference directly within a **React Native** Windows application.
- Implemented the AI services using a **FastAPI** backend, utilizing **OpenAI Whisper** for robust offline speech recognition and the **Intel OpenVINO** toolkit to optimize model inference and kernels on target NPUs.